

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Buisness Analytics
Practical – IX

Roll No :- T053	Name :- Nidhi Yadav
Class :- TYIT	Batch :- 02
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1. Load the dataset & Remove extra spaces from column names

```
import pandas as pd
from scipy.stats import ttest_1samp
df = pd.read_csv("supermarket_sales.csv")
```

✓ 0.0s

```
import pandas as pd
from scipy.stats import ttest_1samp
df = pd.read_csv("supermarket_sales.csv")
df.columns = df.columns.str.strip()
```

✓ 0.0s

2 Display first five records & 3 Select the Total column as Revenue

```
print("First 5 Records:")
print(df.head())
revenue = df["Total"]
```

✓ 0.0s

```
First 5 Records:
Invoice ID Branch City Customer type Gender \
0 750-67-8428 A Yangon Member Female
1 226-31-3881 C Naypyitaw Normal Female
2 631-41-3108 A Yangon Normal Male
3 123-19-1176 A Yangon Member Male
4 173-73-7910 A Yangon Normal Male

Product line Unit price Quantity Tax $K Total Date
0 Health and beauty 74.69 7 26.1415 548.9715 1/5/2019
1 Electronic accessories 35.20 5 3.8200 80.2200 3/8/2019
2 Home and lifestyle 46.33 7 16.2155 340.5255 3/3/2019
3 Health and beauty 58.22 8 23.2880 489.0480 1/27/2019
4 Sports and travel 86.31 7 30.2685 634.3785 2/8/2019

Time Payment cogs gross margin percentage gross income Rating
0 13:00 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4
3 20:33 Ewallet 465.76 4.761905 23.2880 8.4
4 10:17 Ewallet 604.17 4.761905 30.2005 5.3
```

4. Check for missing values & 5 Calculate mean revenue

```
print("\nMissing Values:")
print(revenue.isnull().sum())
revenue = revenue.dropna()
```

Missing Values:
0

```
print("\nMean Revenue:")
print(revenue.mean())
```

✓ 0.0s

Mean Revenue:
322.966749

6. Expected average revenue & 7 Null Hypothesis & 8 Alternative Hypothesis & 9 Significance level 10. Perform one-sample t-test

```
expected_mean = 1500
# H0: μ = 1500
# H1: μ ≠ 1500
alpha = 0.05
t_stat, p_value = ttest_1samp(revenue, expected_mean)
```

✓ 0.0s

11. Display results & 12 & 13. Compare p-value with 0.05 & 14. Business conclusion

```
print("\nT-statistic:", t_stat)
print("P-value:", p_value)
> if p_value < alpha: ...
else:
    print("Null Hypothesis is not rejected")
if p_value < alpha:
    print("The average revenue is significantly different from ₹1500.")
else:
    print("There is not enough evidence to conclude that average revenue differs from ₹1500.")
```

✓ 0.0s

```
T-statistic: -151.37567896776733
P-value: 0.0
Nidhi Yadav T053
Reject Null Hypothesis
The average revenue is significantly different from ₹1500.
```

1 Create the A/B testing dataset data & 2. Display the data & 3. Calculate conversion rate for Group

Part B

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm
data = {
    "Group": ["A", "B"],
    "Users": [1000, 1000],
    "Conversions": [120, 150]
}
df = pd.DataFrame(data)
df.to_csv("ab_testing.csv", index=False)
df = pd.read_csv("ab_testing.csv")
print("A/B Testing Data:")
print(df)
group_a = df[df["Group"] == "A"]
conversion_rate_a = (
    group_a["Conversions"].iloc[0] /
    group_a["Users"].iloc[0]
)
✓ 0.0s
```

A/B Testing Data:

	Group	Users	Conversions
0	A	1000	120
1	B	1000	150

4. Calculate conversion rate for Group B & 5. Compare conversion rates 6. Null Hypothesis 7. Alternative Hypothesis & 8. Significance level & 9. Perform two-proportion test &

```
group_b = df[df["Group"] == "B"]
conversion_rate_b = (
    group_b["Conversions"].iloc[0] /
    group_b["Users"].iloc[0]
)
print("\nConversion Rate A:",
      conversion_rate_a * 100, "%")
print("Conversion Rate B:",
      conversion_rate_b * 100, "%")
if conversion_rate_a > conversion_rate_b:
    print("Group A has a higher conversion rate.")
elif conversion_rate_b > conversion_rate_a:
    print("Group B has a higher conversion rate.")
else:
    print("Both groups have the same conversion rate.")
# H0: pA = pB
# H1: pA != pB
alpha = 0.05
x1 = group_a["Conversions"].iloc[0]
x2 = group_b["Conversions"].iloc[0]
n1 = group_a["Users"].iloc[0]
n2 = group_b["Users"].iloc[0]
p1 = x1 / n1
p2 = x2 / n2
p_pooled = (x1 + x2) / (n1 + n2)
standard_error = (
    p_pooled * (1 - p_pooled) *
    (1/n1 + 1/n2)
) ** 0.5
z_stat = (p1 - p2) / standard_error
p_value = 2 * (1 - norm.cdf(abs(z_stat)))
✓ 0.0s
```

Conversion Rate A: 12.0 %
 Conversion Rate B: 15.0 %
 Group B has a higher conversion rate.

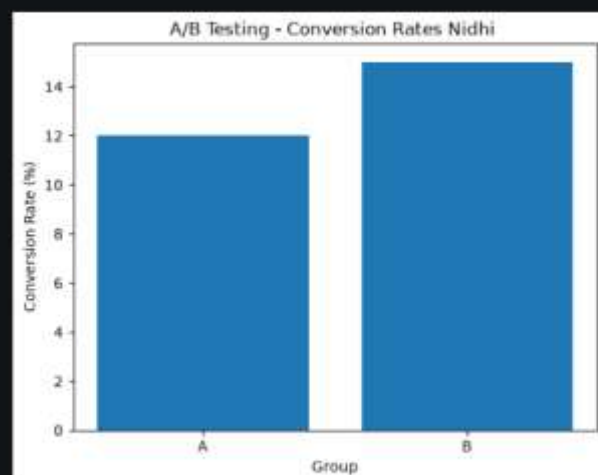
10. Display p-value & 11 & 12. Compare p-value with 0.05 & 13. Create bar chart & 14. Conclusion

```
print("\nZ-statistic:", z_stat)  # "z_stat" is not defined
print("P-value:", p_value)
if p_value < alpha:
    print("Reject Null Hypothesis")
else:
    print("Null Hypothesis is not rejected")

groups = ["A", "B"]

conversion_rates = [
    conversion_rate_a * 100,
    conversion_rate_b * 100  # "conversion_rate_b" is not defined
]
plt.bar(groups, conversion_rates)
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing - Conversion Rates Nidhi")
plt.show()
if p_value < alpha:
    print(
        "\nConclusion: There is a statistically significant "
        "difference between Group A and Group B."
    )
else:
    print(
        "\nConclusion: There is not enough evidence to "
        "conclude that the conversion rates are different."
    )
✓ 0.0s
```

Z-statistic: -1.963849807622355
 P-value: 0.04964038686536876
 Reject Null Hypothesis



Conclusion: There is a statistically significant difference between Group A and Group B.